(1) For a given $\epsilon > 0$, find a $\delta > 0$ such that whenever, $|x - a| < \delta$, we must have $|f(x) - l| < \epsilon$, where $f(x) = 4x-3$, $l = 9$ and $a=3$.

Solution : $|f(x) - l| < \epsilon$

if and only if $|4x-3 - 9| < \epsilon$

if and only if $|4x-12| < \epsilon$

if and only if $|x-3| < \frac{\epsilon}{4}$

Thus required $\delta = \frac{\epsilon}{4}$

2) For a given $\epsilon > 0$, find a $\delta > 0$ such that whenever, $|x - a| < \delta$, we must have $|f(x) - l| < \epsilon$, where $f(x) = 3x-4$, $l = 8$ and $a=4$.

Solution :

$|f(x) - l| < \epsilon$

if and only if $|3x-4-8| < \epsilon$

if and only if $|3x-12| < \epsilon$

if and only if $|3(x-4)| < \epsilon$

if and only if $|x-4| < \frac{\epsilon}{3}$

Thus required $\delta = \frac{\epsilon}{3}$

3) If $x^2-4x+5 \le f(x) \le 2x^2-8x+9$ for all $x \in R$, then find $\lim_{x\to 2} f(x)$.

Solution :

$x^2-4x+5 \le f(x) \le 2x^2-8x+9$

$\lim_{x\to 2}(x^2-4x+5) \le \lim_{x\to 2} f(x) \le \lim_{x\to 2}(2x^2-8x+9)$

$(2)^2-4(2)+5 \le \lim_{x\to 2} f(x) \le 2(2)^2-8(2)+9$

$4-8+5 \le \lim_{x\to 2} f(x) \le 8-16+9$

$1 \le \lim_{x\to 2} f(x) \le 1$ $\therefore \lim_{x\to 2} f(x) = 1$

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4) Evaluate $\lim_{x\to 0}\left[\frac{6^x-3^x-2^x+1}{36^x-9^x-4^x+1}\right]$

Solution : Let $L=\lim_{x\to 0}\left[\frac{6^x-3^x-2^x+1}{36^x-9^x-4^x+1}\right]$

$$L=\lim_{x\to 0}\frac{(3\times 2)^x-3^x-2^x+1}{(9\times 4)^x-9^x-4^x+1}$$

$$L=\lim_{x\to 0}\frac{3^x\times 2^x-3^x-2^x+1}{9^x\times 4^x-9^x-4^x+1}$$

$$L=\lim_{x\to 0}\frac{3^x(2^x-1)-1(2^x-1)}{9^x(4^x-1)-1(4^x-1)}$$

$$L=\lim_{x\to 0}\frac{(9^x-1)(3^x-1)}{(4^x-1)(9^x-1)}$$

$$L=\lim_{x\to 0}\frac{\left(\frac{2^x-1}{x}\right)\left(\frac{3^x-1}{x}\right)}{\left(\frac{4^x-1}{x}\right)\left(\frac{9^x-1}{x}\right)}$$

$$L=\frac{\lim_{x\to 0}\left(\frac{2^x-1}{x}\right)\lim_{x\to 0}\left(\frac{3^x-1}{x}\right)}{\lim_{x\to 0}\left(\frac{4^x-1}{x}\right)\lim_{x\to 0}\left(\frac{9^x-1}{x}\right)}$$

Using $\lim_{x\to 0}\frac{a^x-1}{x}=\log a$

$$L=\frac{(\log 2)(\log 3)}{(\log 4)(\log 9)}$$

$$L=\frac{(\log 2)(\log 3)}{(\log 2^2)(\log 3^2)}$$

$$L=\frac{(\cancel{\log 2})(\cancel{\log 3})}{2(\cancel{\log 2})\,2(\cancel{\log 3})}=\frac{1}{4}$$

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5) Evaluate $\lim_{x\to\frac{\pi}{3}}\left[\frac{2-\cos x-\sqrt{3}\sin x}{(3x-\pi)^2}\right]$

Solution : $L=\lim_{x-\frac{\pi}{3}}\left[\frac{2-\cos x-\sqrt{3}\sin x}{(3x-\pi)^2}\right]$

$$=\lim_{x\to\frac{\pi}{3}}\left[\frac{2-2\left(\frac{1}{2}\cos x+\frac{\sqrt{3}}{2}\sin x\right)}{(3x-\pi)^2}\right]$$

$$=\lim_{x\to\pi/3}\left[\frac{2-2\left(\cos\frac{\pi}{3}\cos x+\sin\frac{\pi}{3}\sin x\right)}{(3x-\pi)^2}\right]$$

$$=\lim_{x\to\pi/3}\left[\frac{2-2\cos\left(x-\frac{\pi}{3}\right)}{(3x-\pi)^2}\right]$$

Put $x-\pi/3=h \Longrightarrow 3x-\pi=3h$

as $\lim: x\to\pi/3,\ h\to 0$

$$\therefore L=\lim_{h\to 0}\left[\frac{2-2\cos h}{(3h)^2}\right]$$

$$L=\lim_{h\to 0}\frac{2(1-\cos h)}{9h^2}$$

$$L=\lim_{h\to 0}\frac{2}{9}\times\frac{2\sin^2 h/2}{h^2}\quad(\because 1-\cos\theta=2\sin^2\theta/2)$$

$$L=\lim_{h\to 0}\frac{4}{9}\times\frac{\sin^2 h/2}{\frac{h^2}{4}}\times\frac{1}{4}$$

$$L=\lim_{h\to 0}\frac{1}{9}\times 1\quad\left(\because \lim_{h\to 0}\frac{\sin\theta}{\theta}=1\right)$$

$$\therefore L=\frac{1}{9}$$

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6) Evaluate $\lim_{x\to\frac{\pi}{2}}\left[\frac{\left(2^{x-\frac{\pi}{2}}-1\right)^3}{\left(3^{x-\frac{\pi}{2}}-1\right)(\cos x)\log\left(\frac{2+2x-\pi}{2}\right)}\right]$

Solution :

$$L=\lim_{x\to\pi/2}\left[\frac{(2^{x-\frac{\pi}{2}}-1)^3}{(3^{x-\frac{\pi}{2}}-1)\cos x\log\left(\frac{2+2x-\pi}{2}\right)}\right]$$

Put $x-\pi/2=h \Rightarrow 2x-\pi=2h$

at $x\to\pi/2,\ h\to 0$

$$\therefore L=\lim_{h\to 0}\left[\frac{(2^h-1)^3}{(3^h-1)\cos\left(\frac{\pi}{2}+h\right)\log\left(\frac{2+2h}{2}\right)}\right]$$

$$L=\lim_{h\to 0}\left[\frac{(2^h-1)^3}{(3^h-1)(-\sin h)\log(1+h)}\right]$$

$$L=\lim_{h\to 0}\left[\frac{\left(\frac{2^h-1}{h}\right)^3}{\left(\frac{3^h-1}{h}\right)\left(\frac{-\sin h}{h}\right)\left(\frac{\log(1+h)}{h}\right)}\right]$$

Using $\lim_{h\to 0}\frac{a^h-1}{h}=\log a$; $\lim_{h\to 0}\frac{\sin\theta}{\theta}$

$\lim_{x\to 0}\frac{\log(1+x)}{x}=1$

$$\therefore L=\frac{(\log 2)^3}{(\log 3)(-1)(1)}$$

$$L=-\frac{(\log 2)^3}{(\log 3)}$$

Sign of Teacher :

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